

JooYoung Seo

PH.D.

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Education

The Pennsylvania State University

University Park, PA

PH.D. IN LEARNING, DESIGN, AND TECHNOLOGY

2021

- Dissertation Title: “Discovering Informal Learning Cultures of Blind Individuals Pursuing STEM Disciplines: A Computational Ethnography Using Public Listserv Archives.”
- Committee members: Drs. Gabriela T. Richard (adviser; dissertation chair), Roy B. Clariana, ChanMin Kim, and Mary Beth Rosson.

The Pennsylvania State University

University Park, PA

M.ED. IN LEARNING, DESIGN, AND TECHNOLOGY

2016

Sungkyunkwan University

Seoul, South Korea

DOUBLE B.A. IN EDUCATION, ENGLISH LITERATURE

2014

Professional Appointments

University of Illinois Urbana-Champaign

Champaign, IL

ASSISTANT PROFESSOR, SCHOOL OF INFORMATION SCIENCES

Sep. 2021 - Present

- Faculty Affiliate: Siebel School of Computing and Data Science (CS), 2024 - present; National Center for Supercomputing Applications (NCSA), 2021 - present; Informatics, 2021 - present; Beckman Institute for Advanced Science and Technology, 2023 - present; Center for Social and Behavioral Science, 2025 - present; IDEA (Inclusion, Diversity, Equity, and Access) Institute, 2022 - present.
- Faculty Theme Lead for Digital Access and Equity, Center for Research and Innovation in Technology-Enhanced Learning (ITEL), College of Education, 2025 - present.

Smith-Kettlewell Eye Research Institute

San Francisco, CA

AFFILIATE SCIENTIST

Jul. 2023 - Present

- Rehabilitation Engineering Research Center (RERC) on Blindness and Low Vision.

RStudio

Boston, MA

SOFTWARE ENGINEER INTERN

May 2020 - Aug. 2020

Awards and Honors

- Center for Advanced Study (CAS) Beckman Fellow, University of Illinois Urbana-Champaign, 2026 - 2027.
- Early Career Award, Institute of Museum and Library Services (IMLS), 2023 - 2026.
- Teach Access Faculty Award, Teach Access, 2023.
- Teachers Ranked as Excellent (four semesters), University of Illinois Urbana-Champaign, 2022 - 2025.
- Best Paper Honorable Mention Award (top 5% of submissions), ACM CHI 2022.
- Emerging Scholar, International Society of the Learning Sciences and the Wallace Foundation, 2021 - 2022.
- Qualcomm Scholarship (Doctoral Consortium), ACM Richard Tapia Conference, 2020.
- NextGen Leader, Disability:IN, 2020.
- Best Doctoral Proposal Fellowship, First International Conference on Quantitative Ethnography, 2019.
- Andrew V. Kozak Memorial Fellowship, PDK Educational Foundation, 2019.
- Delta Gamma Golden Anchor Award, The Pennsylvania State University, 2015 and 2017.
- Global Korean Government Scholarship, National Institute for International Education, 2014 - 2016.
- Outstanding Academic Award, Hyomyoung Scholarship, 2014 - 2019.

Grants

AWARDED GRANTS

- “Collaborative Research: An AudioTactile Data System for Blind or Low Vision Faculty, Staff, Postdocs, and Graduate Students in Chemistry, Math, Computer and Information Sciences,” PI (at UIUC), National Science Foundation (NSF Grant Award #2348166), Workplace Equity for Persons with Disabilities in STEM and STEM Education (nsf23593), \$459,000 (for UIUC) out of \$1,200,159 total, August 2024 - July 2028 (ongoing).
- “Beyond Visuals: Improving Accessibility of Data Curation and Multi-Modal Representations for People of all Abilities through Reproducible Workflows,” PI, Institute of Museum and Library Services (IMLS) Laura Bush 21st Century Librarian Program, Grant #RE-254891-OLS-23, \$649,921, Aug 2023 - July 2027 (early terminated on April 28 and reinstated on May 21, 2025 affected by the President’s Executive Order 14238).
- “Promoting Computational Thinking Skills for Blind and Visually Impaired Teens through Accessible Library Makerspaces,” PI, Institute of Museum and Library Services (IMLS) National Leadership Grants, Grant #LG-252360-OLS-22, \$498,638, Aug 2022 - Mar 2026 (early terminated on April 8th and reinstated on May 21, 2025 affected by the President’s Executive Order 14238).
- “Inclusive Data Science: Fostering Accessibility and Reproducibility with Multimodal Learning Strategies,” PI, Teach Access Faculty Grants, Tier 3, \$5,000, May 2023 - May 2024 (completed).
- “Improving Accessible Reproducibility for Data Science Publishing System,” PI, Posit PBC (unrestricted gift), \$54,327, May 2022 - Present (ongoing).
- “Co-designing an Alexa-Based Conversational mHealth System with Visually Impaired People to Promote Physical Activities,” Co-PI, Center on Health, Aging, and Disability’s (CHAD) Pilot Grant Program at the University of Illinois Urbana-Champaign, \$29,866, 2023-2024 (completed).
- “Category I: Crossing the Divide Between Today’s Practice and Tomorrow’s Science,” Senior Personnel, National Science Foundation (NSF), Grant #2005572, \$10,000,000.00, Oct 2021 - Sept 30, 2026 (.5 FTE academic year) (ongoing).
- “Insight without Sight: Enhancing Data Literacy through Multimodal Data Science Education - Smith-Kettlewell Summer Institute,” Subaward PI, Rehabilitation Engineering Research Centers (RERC) Program: RERC on Blindness and Low Vision, National Institute on Disability, Independent Living, and Rehabilitation Research (NIDILRR), Grant #90REGE0018-01-00, \$11,975, July 17, 2023 - August 11, 2023 (completed).
- “Data Accessibilization: Making Data Science Education Accessible for Blind Learners,” PI, Wallace Foundation, Emerging Scholars Program, International Society of the Learning Sciences, \$10,000, Jan 2022 - Dec 2022 (completed).

PROPOSALS UNDER REVIEW

- “CAREER: Perceptually-Grounded Multimodal AI for Non-Visual Data Reasoning,” PI, National Science Foundation (NSF), Faculty Early Career Development Program (CAREER), \$699,426 requested over five years, submitted July 2026.
- “Collaborative Research: HCC: Conversational Interaction for Non-Visual Chart Sensemaking: Modeling Inquiries, Aligning Models, and Empowering Verification,” PI (at UIUC), National Science Foundation (NSF), Future Computing Research (Future CoRe, NSF 25-543), \$450,000 requested (for UIUC), with Boston College (lead) and the Massachusetts Institute of Technology, submitted September 2026.

Publications

† Student I advised, supervised, or mentored.

REFEREED JOURNAL PAPERS

18. Choi, S., Chen, Y., & **Seo, J.** (2026). Development of an Interactive R Shiny Application for Dynamic Health Disparities Research. *Computers, Informatics, Nursing*, 44(3), e01397. <https://doi.org/10.1097/CIN.0000000000001397>

17. Choi, S., **Seo, J.**, Krishnan, A., Kamath†, S., Hernandez, M. E., & Kitsiou, S. (2026). Managing physical activity through voice: A usability study of the PCHA app with blind and low-vision individuals. *Disability and Rehabilitation: Assistive Technology*, 1–13. <https://doi.org/10.1080/17483107.2026.2663134>
16. Choi, S., **Seo, J.**, Krishnan, A., & Ko, H. S. (2026). Designing an accessible home-based muscle strengthening program for blind and low-vision adults. *Disability and Rehabilitation: Assistive Technology*, 1–12. <https://doi.org/10.1080/17483107.2026.2680547>
15. Choi, S., **Seo, J.**, Krishnan, A., Kamath†, S., Kitsiou, S., & Haegele, J. (2025). Development of a personalized conversational health agent to enhance physical activity for blind and low-vision individuals. *mHealth*, 11, 29. <https://doi.org/10.21037/mhealth-24-60>
14. Choi, G. W., & **Seo, J.** (2024). Accessibility, Usability, and Universal Design for Learning: Discussion of Three Key LX/UX Elements for Inclusive Learning Design. *TechTrends*, 68(5), 936–945. <https://doi.org/10.1007/s11528-024-00987-6>
13. Choi, S., **Seo, J.**, Hernandez, M., & Kitsiou, S. (2024). Conversational agents in mHealth: Use patterns, challenges, and design opportunities for individuals with visual impairments. *Journal of Technology in Behavioral Science*, 9(4), 912–923. <https://doi.org/10.1007/s41347-024-00409-7>
12. Lee, J. G. W., Lee, K., Lee, B., Choi, S., **Seo, J.**, & Choe, E. K. (2023). Personal Health Data Tracking by Blind and Low-Vision People: Survey Study. *Journal of Medical Internet Research*, 25(1), e43917. <https://doi.org/10.2196/43917>
11. Moon, J., Choi, G. W., & **Seo, J. Y.** (2023). Revisiting multimedia learning design principles in virtual reality-based learning environments for autistic individuals. *Virtual Reality*, 27(4), 3101–3113. <https://doi.org/10.1007/s10055-023-00856-2>
10. Moon, J., Lee, D., Choi, G. W., **Seo, J.**, Do, J., & Lim, T. (2023). Learning analytics in seamless learning environments: A systematic review. *Interactive Learning Environments*, 32(7), 3208–3225. <https://doi.org/10.1080/10494820.2023.2170422>
9. **Seo, J.**, & Dogucu, M. (2023). Teaching Visual Accessibility in Introductory Data Science Classes with Multi-Modal Data Representations. *Journal of Data Science*, 21(2), 428–441. <https://doi.org/10.6339/23-JDS1095>
8. **Seo, J.**, & Choi, S. (2022). Are blind people considered scientific knowledge producers? An accessibility report on the top 10 SCIE journal systems using a tripartite evaluation approach. *Journal on Technology and Persons with Disabilities*, 10, 171–183. <http://hdl.handle.net/10211.3/223473>
7. **Seo, J.**, Moon, J., Choi, G. W., & Do, J. (2022). A Scoping Review of Three Computational Approaches to Ethnographic Research in Digital Learning Environments. *TechTrends*, 66(1), 102–111. <https://doi.org/10.1007/s11528-021-00689-3>
6. Choi, S., & **Seo, J.** (2021). Trends in healthcare research on visual impairment and blindness: Use of bibliometrics and hierarchical cluster analysis. *Ophthalmic Epidemiology*, 28(4), 277–284. <https://doi.org/10.1080/09286586.2020.1863993>
5. **Seo, J.**, & Richard, G. T. (2021). SCAFFOLDing All Abilities into Makerspaces: A Design Framework for Universal, Accessible and Intersectionally Inclusive Making and Learning. *Information and Learning Sciences*, 122(11/12), 795–815. <https://doi.org/10.1108/ILS-10-2020-0230>
4. Choi, S., & **Seo, J.** (2020). An exploratory study of the research on caregiver depression: Using bibliometrics and LDA topic modeling. *Issues in Mental Health Nursing*, 41(7), 592–601. <https://doi.org/10.1080/01612840.2019.1705944>
3. Choi, S., & **Seo, J.** (2019). Analysis of caregiver burden in palliative care: An integrated review. *Nursing Forum*, 54(2), 280–290. <https://doi.org/10.1111/nuf.12328>
2. **Seo, J.** (2019). Is the maker movement inclusive of ANYONE?: Three accessibility considerations to invite blind makers to the making world. *TechTrends*, 63(5), 514–520. <https://doi.org/10.1007/s11528-019-00377-3>
1. **Seo, J.**, & McCurry, S. (2019). LaTeX is NOT easy: Creating accessible scientific documents with R Markdown. *Journal on Technology and Persons with Disabilities*, 7, 157–171. <http://hdl.handle.net/10211.3/210398>

FULL PAPERS IN REFEREED CONFERENCE PROCEEDINGS

15. Khan†, O., & **Seo, J.** (2026, October). “I don’t want my mental health app to give me mental health barriers”: Unpacking the need for digital mental health tracking services with and for the Blind community. *Proceedings of the 28th International ACM SIGACCESS Conference on Computers and Accessibility*. <https://doi.org/10.1145/3797867.3829020>. In press.
14. Khan†, O., & **Seo, J.** (2026, October). QUARTZ: Qualitative understanding via accessible representation and visualization. *Proceedings of the 28th International ACM SIGACCESS Conference on Computers and Accessibility*. <https://doi.org/10.1145/3797867.3829019>. In press.

13. Zhang†, W., & **Seo, J.** (2026, October). Learning Python as a Blind Programmer in an AI-Mediated Environment: An Autoethnographic Study. *Proceedings of the 28th International ACM SIGACCESS Conference on Computers and Accessibility*. <https://doi.org/10.1145/3797867.3829037>. In press.
12. Kamath†, S. S., Zeidieh†, A. N., Potluri, V., O'Modhrain, S., Perry, K., & **Seo, J.** (2026). Three modalities, two design probes, one prototype, and no vision: Experience-Based Co-Design of a Multi-modal 3D Data Visualization Tool. *Proceedings of the CHI Conference on Human Factors in Computing Systems*, 1–23. <https://doi.org/10.1145/3772318.3791272>
11. Khan†, O., & **Seo, J.** (2026). “I Don't Trust Any Professional Research Tool”: A Re-Imagination of Knowledge Production Workflows by, with, and for Blind and Low-Vision Researchers. *Proceedings of the CHI Conference on Human Factors in Computing Systems*, 1–24. <https://doi.org/10.1145/3772318.3791242>
10. Kamath†, S. S., Khan†, O., Choudhary†, A., Meyerhoff-Liang, J., Choi, S., & **Seo, J.** (2025). PunchPulse: A Physically Demanding Virtual Reality Boxing Game Designed with, for and by Blind and Low-Vision Players. *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility*, 1–21. <https://doi.org/10.1145/3663547.3746365>
9. Ning, Z., Li, L., Killough, D., **Seo, J.**, Carrington, P., Tian, Y., Zhao, Y., Li, F. M., & Li, T. J.-J. (2025). AROMA: Mixed-Initiative AI Assistance for Non-Visual Cooking by Grounding Multimodal Information Between Reality and Videos. *Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology*, 1–15. <https://doi.org/10.1145/3746059.3747650>
8. Lee, J. G. W., Lee, B., Choi, S., **Seo, J.**, & Choe, E. K. (2024). Identify, Adapt, Persist: The Journey of Blind Individuals with Personal Health Technologies. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 8(2), 51:1–51:21. <https://doi.org/10.1145/3659585>
7. **Seo, J.**, Kamath†, S. S., Zeidieh†, A., Venkatesh†, S., & McCurry, S. (2024). MAIDR Meets AI: Exploring Multimodal LLM-Based Data Visualization Interpretation by and with Blind and Low-Vision Users. *Proceedings of the 26th International ACM SIGACCESS Conference on Computers and Accessibility*, 1–31. <https://doi.org/10.1145/3663548.3675660>
6. **Seo, J.**, O'Modhrain, S., Xia†, Y., Kamath†, S., Lee, B., & Coughlan, J. M. (2024). Designing Born-Accessible Courses in Data Science and Visualization: Challenges and Opportunities of a Remote Curriculum Taught by Blind Instructors to Blind Students. In E. E. Firat, R. S. Laramée, & N. Sindlev Andersen (Eds.), *EuroVis 2024 - education papers*. The Eurographics Association. <https://doi.org/10.2312/eved.20241053>
5. **Seo, J.**, Xia†, Y., Lee, B., McCurry, S., & Yam†, Y. J. (2024). MAIDR: Making Statistical Visualizations Accessible with Multimodal Data Representation. *Proceedings of the CHI Conference on Human Factors in Computing Systems*, 1–22. <https://doi.org/10.1145/3613904.3642730>
4. Kaushik, S., Barbosa, N. M., Yu, Y., Sharma, T., Kilhoffer, Z., **Seo, J.**, Das, S., & Wang, Y. (2023). GuardLens: Supporting safer online browsing for people with visual impairments. *Nineteenth Symposium on Usable Privacy and Security (SOUPS 2023)*, 361–380. <https://www.usenix.org/conference/soups2023/presentation/kaushik>
3. Zhang, Z. (Jerry)., Kaushik, S., **Seo, J.**, Yuan, H., Das, S., Findlater, L., Gurari, D., Stangl, A., & Wang, Y. (2023). ImageAlly: A Human-AI hybrid approach to support blind people in detecting and redacting private image content. *Nineteenth Symposium on Usable Privacy and Security (SOUPS 2023)*, 417–436. <https://www.usenix.org/conference/soups2023/presentation/zhang>
2. Lee, C. Y. P., Zhang, Z., Herskovitz, J., **Seo, J.**, & Guo, A. (2022). CollabAlly: Accessible Collaboration Awareness in Document Editing. *CHI Conference on Human Factors in Computing Systems*, 1–17. <https://doi.org/10.1145/3491102.3517635>. *Best Paper Honorable Mention*.
1. **Seo, J.**, & Richard, G. T. (2020). Coding through touch: Exploring and re-designing tactile making activities with learners with visual dis/abilities. In M. Gresalfi & I. Horn (Eds.), *Interdisciplinarity in the learning sciences, 14th international conference of the learning sciences (ICLS) 2020* (Vol. 3, pp. 1373–1380). International Society of the Learning Sciences (ISLS).

SHORT PAPERS, POSTERS, DEMOS, AND ABSTRACTS

29. Pradhan†, S. K., Anand, A., & **Seo, J.** (2026, October). SK Chess: An accessible chess system for Blind and Low Vision chess players. *Proceedings of the 28th International ACM SIGACCESS Conference on Computers and Accessibility*. In press.
28. Zeidieh†, A., Sharevski, F., Wang, Y., & **Seo, J.** (2026, October). “It's our ride, we're just out of the driver's seat”: Perceptions of and challenges with fully autonomous vehicles among BLV Waymo riders. *Proceedings of the 28th International ACM SIGACCESS Conference on Computers and Accessibility*. In press.

27. Evans, D. J., Naiman, J. P., **Seo, J.**, & Downie, J. S. (2026, September). Do you “see” what I “see”? A multi-panel visual question and answer dataset for large language model chart analysis. *Proceedings of the 30th International Conference on Theory and Practice of Digital Libraries*. In press.
26. Ge†, K., Paul, R., Zhang, P., & **Seo, J.** (2025). Phoenix: A Novel Context-Aware Voice-Powered Math Equation Workspace and Editor. *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility*, 1–6. <https://doi.org/10.1145/3663547.3759766>
25. Kamath†, S. S., Zeidieh†, A. N., & **Seo, J.** (2025). Explore, Listen, Inspect: Supporting Multimodal Interaction with 3D Surface and Point Data Visualizations. *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility*, 1–6. <https://doi.org/10.1145/3663547.3759765>
24. Khan†, O., & **Seo, J.** (2025). “Sighted People Have Their Pick Of The Litter”: Unpacking The Need For Digital Mental Health (DMH) Tracking Services With And For The Blind Community. *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*, 1–13. <https://doi.org/10.1145/3706599.3719817>
23. Koh, K., **Seo, J.**, Chen†, S., & Cox†, E. M. (2025). Engaging with information beyond vision: Hands-on approaches to computational thinking for blind and visually impaired learners. *Information Research an International Electronic Journal*, 30(iConf), 280–286. <https://doi.org/10.47989/ir30iConf47353>
22. Naiman, J. P., Evans, D. J., & **Seo, J.** (2025). What lies beneath: A call for distribution-based visual question & answer datasets. *Proceedings of the 2025 ACM/IEEE Joint Conference on Digital Libraries*, 257–260. <https://doi.org/10.1109/JCDL67857.2025.00040>
21. Zeidieh†, A. N., & **Seo, J.** (2025). Check Now, Can You See It?: Exploring Voice and Video-Capable Language Models for Identifying and Spatially Locating Items of Interest for Blind and Low-Vision Travelers. *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility*, 1–5. <https://doi.org/10.1145/3663547.3749833>. Experience report.
20. Chen†, S., Cox†, E., Koh, K., & **Seo, J.** (2024). The role of verbal descriptions in facilitating computational thinking skills for BVI learners. In R. Lindgren, T. I. Asino, E. A. Kyza, C. K. Looi, D. T. Keifert, & E. Suárez (Eds.), *Proceedings of the 18th international conference of the learning sciences - ICLS 2024* (pp. 2343–2344). International Society of the Learning Sciences. <https://doi.org/10.22318/icls2024.594746>
19. Ge†, K., & **Seo, J.** (2024). StereoMath: An Accessible and Musical Equation Editor. *Proceedings of the 26th International ACM SIGACCESS Conference on Computers and Accessibility*, 1–5. <https://doi.org/10.1145/3663548.3688487>
18. Jung, Y. J., Chang, Y., Moon, J., **Seo, J.**, Bonnette, R., Lee, J.-Y., Ke, F., Sokolikj, Z., Koh, K., Cox†, E., Chen†, S., Abbas, J., Munyao, M., DiCioccio, M., & Alstad, Z. (2024). Learning environments designed for and with learners with disabilities. In R. Lindgren, T. I. Asino, E. A. Kyza, C. K. Looi, D. T. Keifert, & E. Suárez (Eds.), *Proceedings of the 18th international conference of the learning sciences - ICLS 2024* (pp. 1918–1925). International Society of the Learning Sciences. <https://doi.org/10.22318/icls2024.384050>
17. Kamath†, S. S., Zeidieh†, A., Khan†, O., Sethi†, D., & **Seo, J.** (2024). Playing Without Barriers: Crafting Playful and Accessible VR Table-Tennis with and for Blind and Low-Vision Individuals. *Proceedings of the 26th International ACM SIGACCESS Conference on Computers and Accessibility*, 1–5. <https://doi.org/10.1145/3663548.3688526>
16. Zhou, K. Z., Adler, R. F., Almendral, C., Choi, S., Kletenik, D., Oro, B., & **Seo, J.** (2024). Teaching Accessibility in Different Disciplines: Topics, Approaches, Resources, Challenges. *Proceedings of the 26th International ACM SIGACCESS Conference on Computers and Accessibility*, 1–4. <https://doi.org/10.1145/3663548.3688553>
15. Kim, S. H., Yoon, A., & **Seo, J.** (2023). Trend of Collaboration in STEM Education in Informal Learning Institutions Based on IMLS-funded Projects. *Proceedings of the Association for Information Science and Technology*, 60(1), 625–629. <https://doi.org/10.1002/pra2.828>
14. Park†, J., **Seo, J.**, & Lee†, J. Y. (2023). Exploring an Online Community of Blind Programmers by Using Topic Modeling and Network Analysis. *Proceedings of the Association for Information Science and Technology*, 60(1), 1096–1098. <https://doi.org/10.1002/pra2.956>
13. **Seo, J.**, & Rogge, M. (2023). Coding Non-Visually in Visual Studio Code: Collaboration Towards Accessible Development Environment for Blind Programmers. *Proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility*, 1–9. <https://doi.org/10.1145/3597638.3614550>. Experience report.
12. **Seo, J.**, Xia†, Y., Yam†, Y. J., & McCurry, S. (2023). MAIDR: Multimodal Access and Interactive Data Representation System for Inclusive Data Science Education. *Proceedings of the 3rd Annual Meeting of the International Society of the Learning Sciences*, 51–54.

11. Huh†, M., & **Seo, J.** (2022). A duoethnographic study of a mixed-ability team in a collaborative group programming project. In A. Weinberger, W. Chen, D. Hernández-Leo, & B. Chen (Eds.), *Proceedings of the 15th international conference on computer-supported collaborative learning - CSCL 2022* (pp. 471–474). International Society of the Learning Sciences.
10. Lee, C. Y. P., Zhang, Z., Herskovitz, J., **Seo, J.**, & Guo, A. (2021). CollabAlly: Accessible collaboration awareness in document editing. *The 23rd International ACM SIGACCESS Conference on Computers and Accessibility*, 1–4. <https://doi.org/10.1145/3441852.3476562>
9. Choi, S., & **Seo, J.** (2019). Heart failure research using text mining: A systematic review. *NURSING RESEARCH*, 68(2), E119–E120.
8. Choi, S., & **Seo, J.** (2019). Trends in self-management among adults with heart failure from 2011 to 2016 using the national health and nutrition examination surveys. *Journal of Cardiac Failure*, 25(8), S4. <https://doi.org/10.1016/j.cardfail.2019.07.540>
7. Choi, S., **Seo, J.**, & Kitko, L. (2019). Exploring the lived experience of a family member with advanced heart failure: Using a text mining approach. *The First International Conference on Quantitative Ethnography: Poster Session*, S8–S9.
6. **Seo, J.** (2019). Discovering informal learning cultures of blind individuals pursuing STEM disciplines: A quantitative ethnography using listserv archives. *The First International Conference on Quantitative Ethnography: Doctoral Consortium*, S66–S67. *Awarded the best Doctoral Consortium Proposal Cengage fellowship.*
5. Choi, S., & **Seo, J.** (2018). Effects of nonpharmacological interventions for fatigue in patients with heart failure: A systematic review and meta-analysis. *Journal of Cardiac Failure*, 24(8), S73–S74.
4. **Seo, J.** (2018). Accessibility and inclusivity in making: Engaging learners with all abilities in making activities. In *Proceedings of the 3rd learning sciences graduate student conference* (pp. 141–142). LSGSC Planning Team.
3. **Seo, J.**, & Richard, G. T. (2018). Accessibility, making and tactile robotics: Facilitating collaborative learning and computational thinking for learners with visual impairments. In J. Kay & R. Luckin (Eds.), *Rethinking learning in the digital age: Making the learning sciences count, 13th international conference of the learning sciences (ICLS) 2018* (Vol. 3, pp. 1755–1757). International Society of the Learning Sciences (ISLS).
2. Konecki, M., Lovrenčić, S., **Seo, J.**, & LaPierre, C. (2017). The role of ICT in aiding visually impaired students and professionals. *Proceedings of the 11th Multidisciplinary Academic Conference*, 148.
1. **Seo, J.**, AlQahtani, M., Ouyang, X., & Borge, M. (2017). Embracing learners with visual impairments in CSCL. In B. K. Smith, M. Borge, E. Mercier, & K. Y. Lim (Eds.), *Making a difference: Prioritizing equity and access in CSCL, 12th international conference on computer supported collaborative learning (CSCL) 2017* (Vol. 2, pp. 573–576). International Society of the Learning Sciences (ISLS).

BOOK CHAPTERS

1. **Seo, J.**, & Dogucu, M. (2024). Data science + accessibility. In A. Oleson, A. J. Ko, & R. E. Ladner (Eds.), *Teaching accessible computing*. University of Washington. <https://bookish.press/tac/DataScience>

REFEREED CONFERENCE PRESENTATIONS

15. Choi, S., & **Seo, J.** (2025, October). *Advancing health equity for Blind and Low-Vision people through universal design in healthcare* [Conference presentation]. 14th Annual Minority Health Conference.
14. **Seo, J.** (2025, September). *maidr: Empowering accessible, multimodal data visualizations* [Conference presentation]. posit::conf(2025), Atlanta, GA. <https://www.youtube.com/watch?v=QR7mdgM8Hf0>
13. **Seo, J.**, & Dogucu, M. (2022, June). *Teaching visual accessibility in the introductory data science classes: Why, what, when, and how* [Conference presentation]. 2022 Symposium on Data Science & Statistics, Designing Data Science Curricula Session, Pittsburgh, PA.
12. Donegan, S. R., Porter, C., Fogel, A., **Seo, J.**, Choi, S., & Eagan, B. (2021, February). *U.S. media coverage during COVID-19: An epistemic network analysis of bias, topic, and trajectory* [Poster presentation]. 2nd International Conference on Quantitative Ethnography (ICQE), Malibu, CA.
11. **Seo, J.**, & Richard, G. T. (2021, February). *Uncovering latent topics of blind people in computer science: Structural topic modeling for an email corpus* [Poster presentation]. 2nd International Conference on Quantitative Ethnography (ICQE), Malibu, CA.
10. **Seo, J.** (2021, January). *Accessible data science beyond visual models* [Conference presentation]. rstudio::global(2021), Virtual. <https://posit.co/resources/videos/accessible-data-science-beyond-visual-models-non-visual-interactions-with-r-and-rstudio-packages/>
9. **Seo, J.**, & Richard, G. T. (2020, April). *Maker inclusivity = maker accessibility: Further interrogations for diverse participation* [Poster presentation]. Annual Meeting of the American Educational Research Association (AERA), Virtual.

8. **Seo, J.**, & Richard, G. T. (2018, April). *Furthering inclusivity in making: A framework for accessible design of makerspaces for learners with disabilities* [Poster presentation]. Annual Meeting of the American Educational Research Association (AERA), New York City, NY.
7. Bunag, T., Aniela, L., Nielsen, M. C., & **Seo, J.** (2017, November). *The rapidly changing world of accessible online learning* [Panel discussion]. DDL - Accessible Online Learning, Concurrent Presentation of the Association for Educational Communications and Technology (AECT), Jacksonville, FL.
6. **Seo, J.** (2017, September). *Tactile access to visualized statistical data using R* [Poster presentation]. Annual Meeting of the ACM Richard Tapia Celebration of Diversity in Computing, Atlanta, GA.
5. Liao, J., Patcyk, M., **Seo, J.**, & Hooper, S. (2016, October). *Using hierarchical linear modeling to measure growth rate in a gamified CBM environment* [Paper presentation]. Annual Meeting of the Northeastern Educational Research Association (NERA), Trumbull, CT.
4. Kim, K., **Seo, J.**, & Clariana, R. B. (2016, March). *Automatic knowledge structure measure in online courses* [Conference presentation]. Annual Meeting of the Teaching and Learning with Technology (TLT) Symposium, University Park, PA.
3. **Seo, J.** (2016, March). *Engaging blind learners in statistics study using R* [Conference presentation]. Annual Meeting of the Teaching and Learning with Technology (TLT) Symposium, University Park, PA.
2. **Seo, J.** (2015, November). *Assistive technologies for equal access in general education* [Conference presentation]. Annual Meeting of the Association for Educational Communications and Technology (AECT), Indianapolis, IN.
1. **Seo, J.**, & Park, E. (2015, October). *The more accessible, the more potential: Simple tips for online accessibility* [Conference presentation]. Technology and Learning Conference, Blue Bell, PA.

Invited Talks and Presentations

KEYNOTES, INVITED TALKS, AND PANELS

Seo, J. (2024, December). *MAIDR Demo*. Invited demonstration for the MathWorks Accessibility Customer Advisory Board, virtual.

Seo, J. (2024, December). *Making Data Visualization Accessible*. Invited talk at the Research IT Group, University of Illinois Urbana-Champaign, Champaign, IL (virtual).

Invited featured demo (2024, March): *GitHub Copilot Vision Accessibility*. “Building Inclusive Experiences with GitHub Copilot,” Microsoft Accessibility Summit: <https://youtu.be/INhRIgGE4w?feature=shared&t=259>

Seo, J. (2024, March). Invited keynote speaker at the 16th Annual College of Education Graduate Student Conference, “Education in the 21st Century: Designing a More Inclusive Future,” University of Illinois Urbana-Champaign, Champaign, IL.

Seo, J. (2024, March). *Unseen Insights: Empowering Autonomous Data Science by, with, and for Blind People*. Invited talk at the UC Irvine Donald Bren School of Information & Computer Sciences, Irvine, CA.

Seo, J. (2023, November). *Insight Over Eyesight: My Journey as a Blind Data Scientist*. Invited talk at the Chicago Lighthouse Career Fair, Chicago, IL (virtual).

Seo, J. (2023, November). *Beyond Sight: Envisioning Data Science Through Multimodal Access*. Invited as a keynote speaker at the 2023 UIUC iSchool’s Research Showcase, Champaign, IL.

Seo, J. (2023, July). *Screen Reader User Experience Deep Dive*. Invited talk at the Microsoft Visual Studio Code Team, virtual.

Seo, J. (2023, June). *Perceiving Beyond Vision: My Journey Through Dreamscapes, Numerical Cognition, and a Blind Critique of Inclusive Data Visualization*. Invited talk at the Dagstuhl Inclusive Data Visualization Workshop, Dagstuhl, Germany.

Seo, J. (2023, May). *Enhancing Accessibility in Programming and Fostering Inclusive Data Science via Multimodal Strategies*. Invited talk at the Campus Research Computing Consortium, virtual.

Invited panel (2023, May): *Human Computer Relationships @Work: Embodiment, Inclusive Design, Accessibility, and Emerging Technologies for Learning*, hosted by PhET Interactive Simulations at University of Colorado Boulder.

Invited demo by Microsoft (2023, April): *VSCoDe Accessibility*. VS Code Day 2023: <https://m.youtube.com/watch?v=X3qUf0B3x8Q>

Invited panel (2022, December): *Integrating Accessibility into a Computing Career*, hosted by Associate Professor Stacy Branham at Informatics of the University of California, Irvine. (virtual).

Seo, J. (2022, September). *Digital Accessibility*. Invited panel at the INFOcon 2022, virtual.

Seo, J. (2022, April). *Accessible Data Visualization in a Reproducible Way*. Invited talk at Champaign-Urbana Data Science User Group, Champaign, IL (virtual).

Seo, J. (2021, October). *Non-Visual Strategies for Making Statistical-Computing Accessible*. Invited lightning talk at the National Center for Supercomputing Applications, Champaign, IL (virtual).

Seo, J. (2021, September). *Accessibility of Science Beyond Content Accessibility*. Invited talk at the Argonne National Laboratory (Maths and Computer Science Division) (virtual): <https://www.anl.gov/event/accessibility-of-science-beyond-content-accessibility>.

Invited panel (2021, August). DO-IT Neuroscience for Neurodiverse Learners Faculty panel at the University of Washington, virtual.

Seo, J. (2021, June). *How to Learn to Code*. Invited Talk presented at the Nature, Webcast: <https://doi.org/10.1038/d41586-021-01638-z>.

GUEST LECTURES

Seo, J. (2024, April). Invited guest speaker at EPSY 203: Social Issues Group Dialogues (Section AD7: Disability Social Justice), University of Illinois Urbana-Champaign, Champaign, IL.

Seo, J. (2023, November). *Multimodal Data Representation for Inclusive Data Science Education*. Invited talk at the UIUC CI590 DELTA Seminar, Champaign, IL.

Seo, J. (2022, November). *Accessible Library Makerspaces*. Invited talk at the UIUC iSchool IS571: Accessibility in Library Resources, Champaign, IL (virtual).

Seo, J. (2022, March). *Understanding Multi-Modality of Blind People Towards Inclusive STEM Learning*. Invited talk at the UIUC iSchool Proseminar, Champaign, IL (virtual).

Seo, J. (2022, March). *Universal Design = Usability * Accessibility (With Practical Examples Beyond Conceptual Frameworks)*. Invited talk at INTE 535 Analysis, Implementation, and Evaluation of Assistive Technology. Instructional Technology program of the University of Alabama, Tuscaloosa, AL (virtual).

Seo, J. (2022, March). *Your Condition is Not Your Conclusion*. Invited talk at HTM4311: Hotel Professional Development at the Hong Kong Polytechnic University (virtual).

Seo, J. (2021, November). *SCAFFOLDing All Abilities into Makerspaces*. Invited talk at the UIUC iSchool IS582: Make, Design, & Learn in Libraries, Champaign, IL (virtual).

Media Coverage

- Katsnelson, A. (2023, March 6). These tools help visually impaired scientists read data and journals. *Nature*, 615(7951), 362–363. <https://doi.org/10.1038/d41586-023-00645-6>
- Heckel, J. (2023, September 29). *Information sciences professor developing tool to make data visualizations accessible to blind researchers, students*. News Bureau University of Illinois Urbana-Champaign. <https://news.illinois.edu/view/6367/107909242>
- McCallen, B. (2023, November 3). *Data visualization for the blind*. <https://speakingoutfortheblind.weebly.com/list-of-episodes-and-show-news/for-more-information-episode-329-data-visualization-tool-for-the-blind>

Software Developments and Publications

SOFTWARE PACKAGES

- **Seo, J.**, & Kalaiselvan†, N. (2026). *maidr: Multimodal access and interactive data representation* (R package version 0.5.0) [Computer software]. <https://CRAN.R-project.org/package=maidr>. Over 3K downloads.

- **Seo, J., & Venkatesh†, S. (2024).** *maidr: Multimodal access and interactive data representations* (Python package version 1.26.0) [Computer software]. <https://pypi.org/project/maidr/>. Over 86K downloads.
- **Seo, J. (2023).** *maidr: Multimodal access and interactive data representation* (JavaScript package version 4.14.0) [Computer software]. <https://www.npmjs.com/package/maidr>. Over 35K downloads.
- Iannone, R., Cheng, J., Schloerke, B., Haughton, S., Hughes, E., Lauer, A., François, R., **Seo, J.**, Brevoort, K., & Roy, O. (2026). *gt: Easily create presentation-ready display tables* (R package version 1.3.0) [Computer software]. <https://CRAN.R-project.org/package=gt>. Over 7.9M downloads.
- **Seo, J., & Choi, S. (2020).** *youtubecaption: Downloading YouTube subtitle transcription in a tidy tibble data frame* (R package) [Computer software]. <https://CRAN.R-project.org/package=youtubecaption>. Over 18K downloads; archived on CRAN in 2023.
- **Seo, J., & Choi, S. (2019).** *mboxr: Reading, extracting, and converting an mbox file into a tibble* (R package) [Computer software]. <https://CRAN.R-project.org/package=mboxr>. Over 26K downloads; archived on CRAN in 2023.
- **Seo, J., & Choi, S. (2019).** *ezpickr: Easy data import using GUI file picker and seamless communication between an Excel and R* (R package) [Computer software]. <https://CRAN.R-project.org/package=ezpickr>. Over 35K downloads; archived on CRAN in 2022.

OPEN-SOURCE PROJECTS ON GITHUB

- The maidr ecosystem (the TypeScript, Python, and R packages above) is developed in the open under the (x)Ability Design Lab organization: <https://github.com/xability>
- **Seo, J. (2020).** *edmdown: Writing a reproducible article for Journal of Educational Data Mining in R Markdown* (R package version 0.0.0.9000) [Computer software]. <https://github.com/jooyoungseo/edmdown>
- **Seo, J., Chan, T., Michaelis, J. E., & Rosenberg, J. M. (2020).** *islsdown: Writing a reproducible conference paper for the International Society of the Learning Sciences annual meeting in R Markdown* (R package version 0.0.0.9000) [Computer software]. <https://github.com/jooyoungseo/islsdown>
- **Seo, J., & Rosenberg, J. M. (2020).** *jlardown: Writing a reproducible article for Journal of Learning Analytics in R Markdown* (R package version 0.0.0.9000) [Computer software]. <https://github.com/jooyoungseo/jlardown>
- **Seo, J., & McCurry, S. (2019).** *AROW: Accessible RMarkdown online writer* [Computer software]. <https://github.com/ellvix/AROW>
- **Seo, J. (2019).** *ezviewr: View tidy data in preferable spreadsheet* (R package version 0.1.0) [Computer software]. <https://github.com/jooyoungseo/ezviewr>
- **Seo, J. (2019).** *tactileR: Converting R graphics into a braille ready-to-print PDF* (R package version 0.1.0) [Computer software]. <https://github.com/jooyoungseo/tactileR>

MERGED CODE CONTRIBUTIONS TO OPEN-SOURCE PROJECTS

My GitHub pull requests have been peer-reviewed and merged into active open-source data science projects:

- Quarto, Visual Studio Code extension (Posit PBC, 2023): added diagram executable cell engines (quarto-dev/quarto #92).
- quarto-webr, Quarto extension for WebR (2023): improved accessibility by replacing the CodeMirror editor with the Monaco code editor (coatless/quarto-webr #16).
- vscode-R, R extension for Visual Studio Code (2022): contributed R and R Markdown snippets (vscode-R #1009, #1012, #1084, and #1085).
- rticles, article formats for R Markdown (2021): added the reproducible Journal of Educational Data Mining article template (rticles #251; released in v0.21).
- shiny, web application framework for R (2020): contributed to improving keyboard and TTS accessibility for input widgets.

- vitae, curriculum vitae for R Markdown (2020): contributed to enabling LaTeX engine customization and to displaying multiple bibliographies based on a Lua filter.
- rmarkdown, dynamic documents for R (2019): contributed to improving ioslides_presentation accessibility for screen reading software.
- bookdown, authoring books and technical documents with R Markdown (2019): developed nine output-format functions (context_document2, github_document2, beamer_presentation2, and six others) that let R Markdown users cross-reference figures and tables.
- distill, R Markdown format for scientific and technical writing (2019): improved screen reader accessibility for the navigation bar; contributed to displaying more metadata for journal article entries; added missing Google Scholar meta tags.
- wordcountaddin, word counts and readability statistics in R Markdown documents (2019): contributed to the text_stats() function, which gives a word count and other statistics for selected text.
- BrailleR, improved access for blind users (2019): contributed to the BRLThis() function, which lets blind R users convert a graph into a PDF ready for embossing in braille.

Teaching Experience

COURSES TAUGHT (UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN)

- IS 204: Research Design for Information Sciences (online, 3 credit hours): Spring 2026.
- IS 568: Accessibility in HCI Seminar (graduate seminar, 4 credit hours; newly developed): Spring 2025.
- IS 407: Introduction to Data Science (4 credit hours): Spring 2022 (in person); Fall 2022, Spring 2023, Fall 2023, Spring 2024, and Fall 2024 (online).
- IS 226: Introduction to Human-Computer Interaction (online, 3 credit hours; co-taught with Dr. Yang Wang): Fall 2021.
- Teacher Ranked as Excellent by Their Students, Center for Innovation in Teaching and Learning: Spring 2022, Spring 2024, Fall 2024, and Spring 2025.

CURRICULUM DEVELOPMENT

- Designed and launched IS 568: Accessibility in HCI Seminar, a new graduate course (Spring 2025).
- Developed accessibility modules for IS 226: Introduction to Human-Computer Interaction (Fall 2021) and a GitHub-organization-based workflow for IS 407: Introduction to Data Science (Spring 2022).

PROFESSIONAL DEVELOPMENT TEACHING

- Non-Visual Data Science Course for Blind Individuals, Smith-Kettlewell Eye Research Institute (Summer 2023, 2024, and 2025): organized and delivered professional development on non-visual data science for blind individuals: <https://www.ski.org/projects/empowering-data-vision-data-science-course-blind-individuals>

PHD ADVISING

Current advisees

- Sanchita Kamath (Information Sciences), Fall 2023 - present; passed the field exam in Spring 2025.
- Aziz Zeidieh (Informatics), Fall 2023 - present; passed the field exam in Spring 2025.
- Omar Khan (Computer Science), Fall 2024 - present; passed the qualifying exam in Spring 2025.
- Weijun Zhang (Information Sciences), Fall 2025 - present.

PhD examination committees (non-advisees)

- Alice Shinhye Cha (Information Sciences), field exam committee, Spring 2026 - present (advisor: Kyungwon Koh).
- Seung Heon Lee (Computer Science), qualifying exam committee, Fall 2025 (advisor: Koustuv Saha).

- Yilin Xia (Information Sciences), preliminary exam committee, Fall 2025 - present (advisors: Bertram Ludäscher and Matthew Turk).
- Christopher Perdriau (Computer Science), preliminary and final defense committees, Spring 2025 - present (advisor: Colleen Lewis).
- HaeJin Lee (Information Sciences), field exam, preliminary, and final defense committees, Spring 2024 - present (advisor: Nigel Bosch).
- Jaihyun Park (Information Sciences; now Assistant Professor at Indiana University Indianapolis), field exam through final defense, Spring 2022 - Spring 2024 (advisor: Ryan Cordell).

Other supervised PhD researchers (research assistants)

- Si Chen (Information Sciences), research assistant, Spring 2023 - Summer 2024.
- Yilin Xia (Information Sciences), research assistant, Summer 2023.
- Eugene Cox (Informatics), research assistant, Summer 2023 - Summer 2024.

MASTER'S AND UNDERGRADUATE RESEARCH SUPERVISION

Master's research assistants

- Saairam Venkatesh (MS CS; now Software Engineer at Amazon), research assistant, Spring 2024 and Spring 2025.
- Daksh Pokar (MS CS; now Software Engineer at Meta), research assistant, Fall 2024 - Spring 2025.
- Anurag Choudhary (MS CS), research assistant, Fall 2024 - Spring 2025.
- Krishna Anandan Ganesan (MS CS), research assistant, Fall 2024.
- Pritham Sriram Govindaraj (MS CS), graduate assistant for the NSF-funded Delta HPC accessibility project at NCSA, Spring 2023 - Summer 2024.
- Sristi Shivanand Ingleshwar (MS CS), research assistant on the CHAD pilot grant, Fall 2023.
- Krishna Jayachandran (M.Arch), research assistant on the IMLS project, Fall 2022.
- Yash Prakash Bajaj (MS Statistics), graduate assistant for the NSF-funded Delta HPC accessibility project at NCSA, Spring - Summer 2022.

Undergraduate researchers

- Jack Bai (BS Computer Engineering; now UIUC MS CS student), NCSA SPIN (Students Pushing INnovation) program, Summer 2022.
- Yu Jun Yam (BS CS), research internship (Summer 2022) and undergraduate hourly researcher (Fall 2022 - Spring 2023).

Independent studies and research practica supervised

- PhD students: Yilin Xia, Jaihyun Park, Si Chen, HaeJin Lee, Jason Situ, Omar Khan, Aziz Zeidieh, Sanchita Kamath, Alice Shinhye Cha, and Olivia Wang (2022 - 2026).
- Master's students: Chetna Beri, Ankita Khiratkar, Aastha Jain, and Rebekah Chuang (MSIM); Nathaniel Knize (MSLIS); Elle Olivia Dietz (M.Arch); Anurag Choudhary, Daksh Pokar, Aarthi Kesavan Padmanaban, Saairam Venkatesh, Bhoomika Narasimhulu, Akash Elumalai, Niranjan Kalaiselvan, and Soundarya Pradhan (MS CS) (2022 - 2025).
- Undergraduate students: Miso Kim, Khushi Jain, Riya Shah, Harish Ravisankar, Sky Liu, and Jae Young Lee (BSIS); Wenhao (Leo) Zhang, Yu Jun Yam, and Minhyung Lee (BS CS) (2022 - 2023).

EXTERNAL MENTORING

- Kenny Ge (BS CS, Carnegie Mellon University; now Research Intern Fellow at Anthropic), accessibility research mentoring, Fall 2023 - present.

- Mina Huh (BS CS, KAIST; now EECS PhD student at UC Berkeley), accessibility research mentoring, Fall 2021 - Summer 2022; led to a CSCL 2022 paper (Huh & Seo).
- Dhruv Sethi (BTech CS, Manipal Academy of Higher Education, Dubai), internship and professional training, Spring 2024.
- Krishna Jayachandran and Sujithra Rajan (UIUC MS CS graduates), voluntary internships in accessible computing, Spring 2025.
- Ji Yeon (Ariel) Han (PhD student, UC Irvine Informatics) and Jullia Lim (PhD candidate, Teachers College, Columbia University), Korean Edutech/Learning Sciences Researcher Network (KELS) mentoring program, Spring 2023 and Fall 2024.

Service

PROFESSIONAL SERVICE

- Review Panelist, National Science Foundation (NSF), Division of Computer and Network Systems (CISE/CNS), 2025.
- AccessComputing Partner, NSF-funded AccessComputing Alliance, University of Washington, Feb. 2024 - Present: contributing to broadening participation of people with disabilities in computing through mentoring, curriculum sharing, and outreach.
- Program Committee Member: ACM ASSETS 2024; International Conference on Quantitative Ethnography (ICQE) 2023; International Web for All Conference (W4A) 2022.
- Accessibility Chair, 15th International Conference on Educational Data Mining (EDM), Oct. 2021 - Jul. 2022.
- Conference Reviewer: ACM CHI (2020, 2021, 2022, 2023, 2025); ACM ASSETS (2023); ACM IDC (2026); AERA SIG-Learning Sciences (2023); EDM (2022); W4A (2022); the Network of Academic Programs in the Learning Sciences (NAPLeS) of the International Society of the Learning Sciences (2020); ICLS (2019); ICQE (2019).
- Journal Reviewer: IEEE Computer Graphics and Applications (2025); IEEE Security & Privacy (2024); Information and Learning Sciences (2023); TechTrends, special issue on ethnography and learning design (2021).

ADVISORY ROLES

- Accessibility Customer Advisory Board Member, MathWorks, 2023 - Present: providing feedback on MATLAB accessibility features, generative AI integration, and assistive technology compatibility.
- Accessibility Advisory Member and Co-Designer, Microsoft Visual Studio Code (VS Code) Accessibility Team, 2022 - Present: collaborating on screen reader support, terminal buffer, semantic audio cues, Copilot, and Jupyter Notebook accessibility.
- Advisory Board Member, "Making is for Everyone: Learning within Library Makerspaces for Youth with Disabilities" (IMLS National Leadership Grant #LG-252323-OLS-22), 2022 - Present.
- Accessibility Consultant (Faculty Affiliate and Senior Personnel), NSF-funded Delta GPU High-Performance Computing Project, National Center for Supercomputing Applications (NCSA), Fall 2022 - Present.

UNIVERSITY SERVICE (UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN)

- Digital Accessibility and Excellence Steering Committee, Office of the Provost, Spring 2024 - Present (appointed by the University Provost).
- Proposal Reviewer, Chancellor's Call to Research Program (full proposals), Fall 2024.
- Panel Moderator, Disability Etiquette panel at the Access Illinois event, Spring 2023.
- School of Information Sciences: Doctoral Studies Committee (Fall 2023 - Spring 2026); Search Committee for Tenure-Track Information Accessibility Faculty (Fall 2022 - Spring 2023); Data Science Task Force Committee (Fall 2022); Search Committee for Director/Assistant Dean, Diversity, Equity, Inclusion, and Accessibility (Apr. 2022 - Aug. 2022); Diversity Committee (Fall 2021 - Spring 2023).